

Computer Networks (2025-2026)

Part 5: Link Layer

Jeroen Famaey

jeroen.famaey@uantwerpen.be

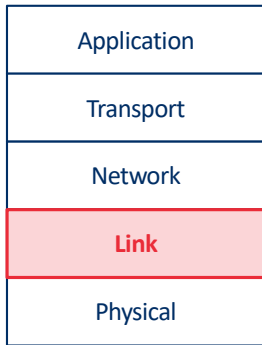
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Introduction to the Link Layer



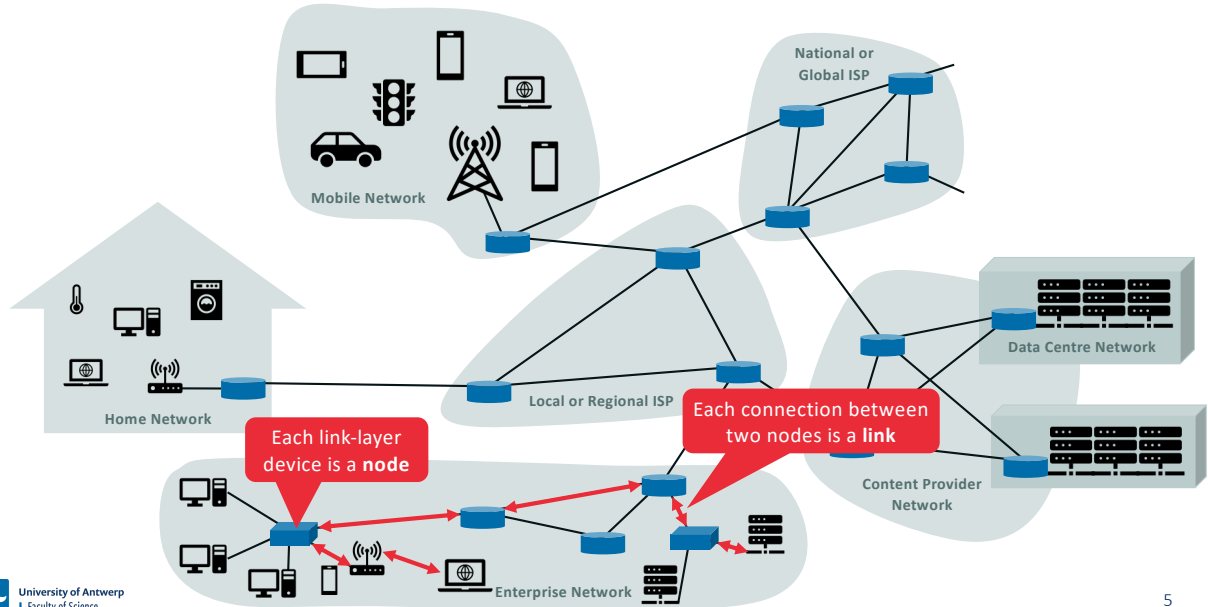
Link layer overview



Link Layer

- Responsible for transferring data across a single link (i.e., between 2 nodes)
- Link-layer packets are called **frames**
- Functionality depends on the specific link-layer protocol
 - Reliability
 - Error detection
 - Addressing
 - Media access
 - ...
- Examples
 - Ethernet
 - Wi-Fi
 - DOCSIS (Data Over Cable Service Interface Specification)
 - PPP (Point-to-Point Protocol)

The link layer enables transmission of data over a single link



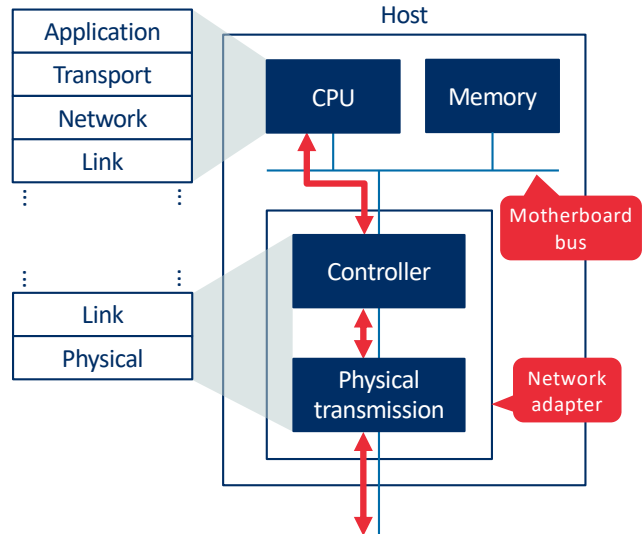
Link layer services

Possible functionality offered by different link layer protocols:

- **Framing**: Encapsulate each network-layer datagram in a link-layer frame (including header fields)
- **Link access**: Medium access control (MAC) protocol that specifies the rules of frame transmissions
- **Reliable delivery**: Guarantee to move each network-layer datagram across the link error-free
- **Error detection and correction**: Detecting and/or correcting bit errors due to attenuation or noise

Link layer implementation

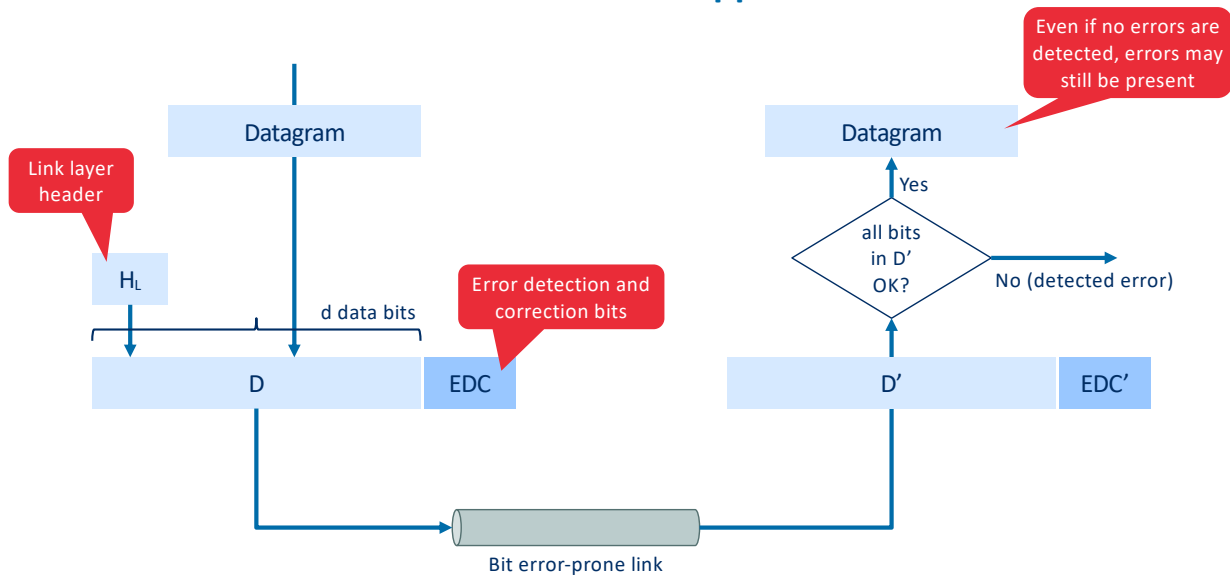
- Many link layer functionalities implemented in network adapter hardware:
 - Framing
 - Error detection
 - ...
- Higher-level link-layer functionality implemented in software:
 - Link-layer addressing
 - Activating the network adapter
 - Frame reception interrupts
 - Handling error conditions
 - ...



Error-detection and -correction



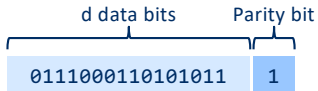
General error-detection and –correction approach



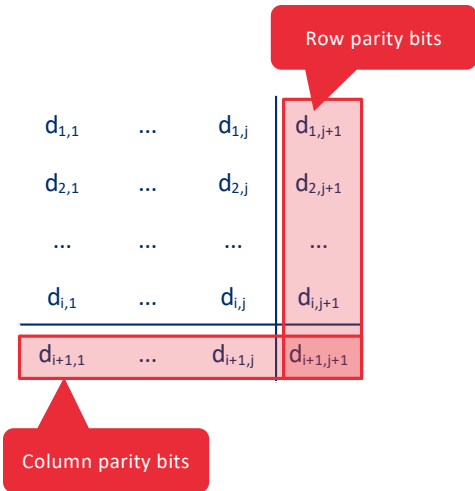
Parity checks

Parity bit

- Add a single error detection bit so that the sum of all $d + 1$ bits is always even (or odd)
- It can only detect an odd number of bit errors (an even number of errors would go undetected)
- Doesn't work well in practice, as bit-errors often occur in bursts (i.e., they are not independent)
- Example (even parity scheme):



Two-dimensional generalization of parity scheme



- The d bits in D are divided in i rows and j columns
- A parity value is computed for each row and column
- Single bit errors will show an error in both the row and column parity bit, and can thus be corrected
 - The ability to correct errors is known as **Forward Error Correction (FEC)**

Exercise: Correcting errors using the two-dimensional parity scheme



Exercise: Find and correct the single bit-error in the below datagram D'.

- Received datagram D' and row/column parity bits (assuming **even** parity checks):

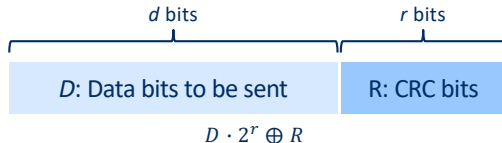
						Row parity bits ↓
	1	0	1	0	1	1
	1	0	1	1	0	0
	0	1	1	1	0	1
Column parity bits →	0	0	1	0	1	0

Checksumming

- The d bits of data are split into k -bit integers, and the sum of these integers is used for error detection
- The **Internet checksum** is based on this approach (RFC 1071)
- As discussed in Part 3, checksums have a lot of limitations, but are easy to calculate (in software)
- A brief reminder on the Internet checksum approach:
 - Sender calculates the 1s complement of the sum of all 16-bit words in the UDP/TCP segment (+ pseudo IP header)
 - Receiver adds all 16-bit words of the segment, pseudo header and checksum together and checks if it is 11111111 11111111
 - At network layer, IP does a similar check, but only on the IP header

Cyclic redundancy check (CRC) codes

- Widely used in today's computer networks (and especially link layer protocols)
- Based on polynomial arithmetic
 - Each bit string can be represented as a polynomial with binary coefficients (e.g., 1011 equals x^3+x+1)
 - All calculations are done in modulo-2 arithmetic without carry in addition or borrows in subtraction
 - Addition and subtraction are thus identical and equal to the bitwise exclusive-OR (XOR, or $\oplus$) operation
 - Multiplication by 2^k can be calculated by left shifting a bit pattern by k places
- Approach
 - Sender and receiver agree on an $r + 1$ bit pattern, known as a **generator G**
 - Sender appends r bits R to the data D so that the resulting $d + r$ bit pattern is divisible by G
 - The receiver divides the $d + r$ received bits by G , if the remainder is non-zero an error has occurred



CRC: How does the sender compute R?

- The $d + r$ bit pattern should be divisible by G , or:

$$D \cdot 2^r \oplus R = nG$$

- If we XOR both sides with R , we get:

$$D \cdot 2^r = nG \oplus R$$

- We can thus calculate R as:

$$R = \text{remainder} \left(\frac{D \cdot 2^r}{G} \right)$$

Exercise: Calculate the CRC code R for a given datagram D



Exercise: Calculate the CRC bits R for $D = 101110$, $G = 1001$, and $r = 3$.

Given:

$$R = \text{remainder} \left(\frac{D \cdot 2^r}{G} \right)$$

Homework: CRC validation



Exercise: Assume $r = 3$, $G = 1001$, and a 9 bit message (consisting of $D + R$) 101110011 is received. Did a bit error occur during transmission?

CRC standards

- Standards have been defined for 8-, 12-, 16-, and 32-bit CRC generators
- Several IEEE link layer protocols use the CRC 32-bit standard with generator

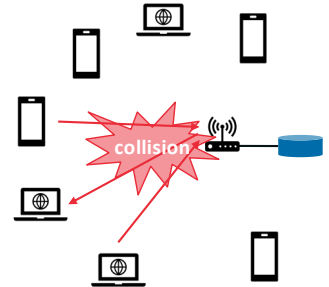
$$G_{\text{CRC-32}} = 100000100110000010001110110110111$$

- Error detection probability of the standardized CRC generators
 - Can detect up to $r - 1$ consecutive bit errors
 - Can also detect consecutive bit errors of length greater than $r + 1$ with a probability equal to $1 - 0.5^r$
 - Can detect any odd number of bit errors

Multiple access links and protocols

The multiple access problem

- When many nodes transmit and receive data over a shared broadcast medium, a **multiple access protocol** is needed to avoid **collisions** (i.e., multiple transmissions at the same time)
- Many types of networks make use of a broadcast medium (e.g., cable access networks, Wi-Fi, satellite)
- Multiple access protocols can be classified in three categories:
 - Channel partitioning protocols
 - Random access protocols
 - Taking-turns protocols



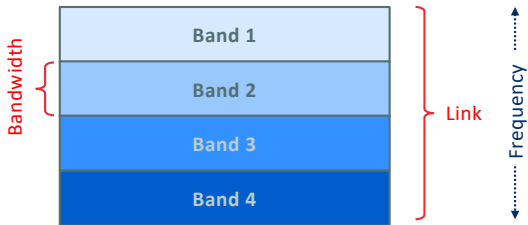
Desirable characteristics of multiple access protocols

Given a broadcast channel of rate R bits per second (bps), a multiple access protocol should have the following desirable characteristics:

1. When only one node has data to send, that node has a throughput of R bps
2. When M nodes have data to send, each should have an average throughput of R / M bps
3. The protocol is decentralized (i.e., there is no master node that acts as a single point of failure)
4. The protocol is simple and inexpensive to implement

Channel partitioning protocols: FDMA and TDMA

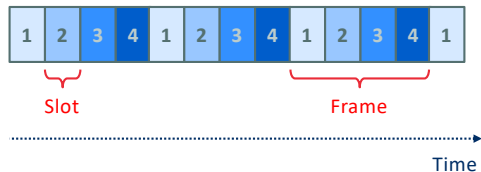
Frequency Division Multiple Access (FDMA)



Each frequency band is dedicated to a specific sender-receiver pair

Orthogonal FDMA (OFDMA)
is used in 4G, 5G, and Wi-Fi 6

Time Division Multiple Access (TDMA)



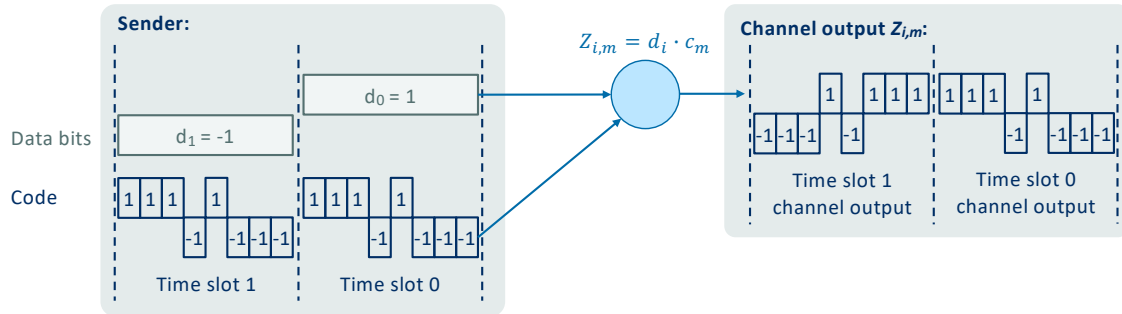
All slots with the same label are dedicated to a specific sender-receiver pair

TDMA is used in 2G mobile networks and DECT

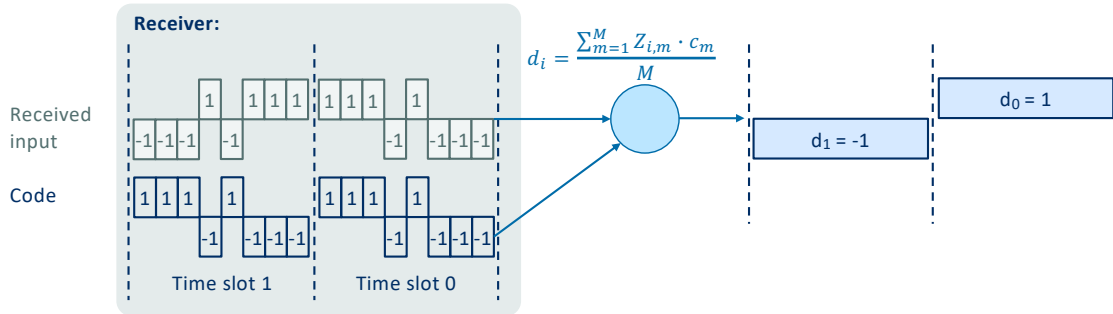
Channel partitioning protocols: CDMA

Code Division Multiple Access (CDMA)

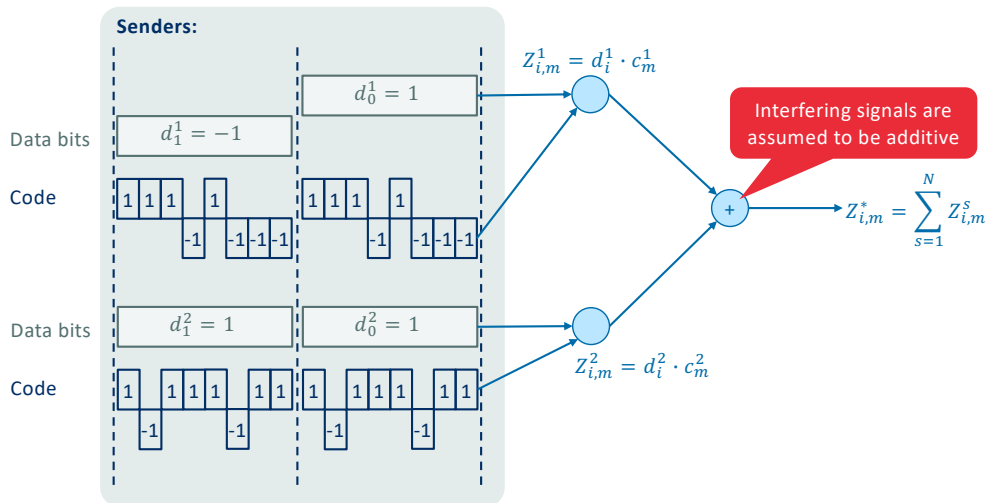
- Each sender is assigned a unique M-bit code
- The code changes at a much faster rate than the sequence of data bits (i.e., the chipping rate)
- Each transmitted bit is encoded by multiplying it by the code
- Example (for convenience, a value of -1 is used for 0 bits):



Channel partitioning protocols: CDMA receiver



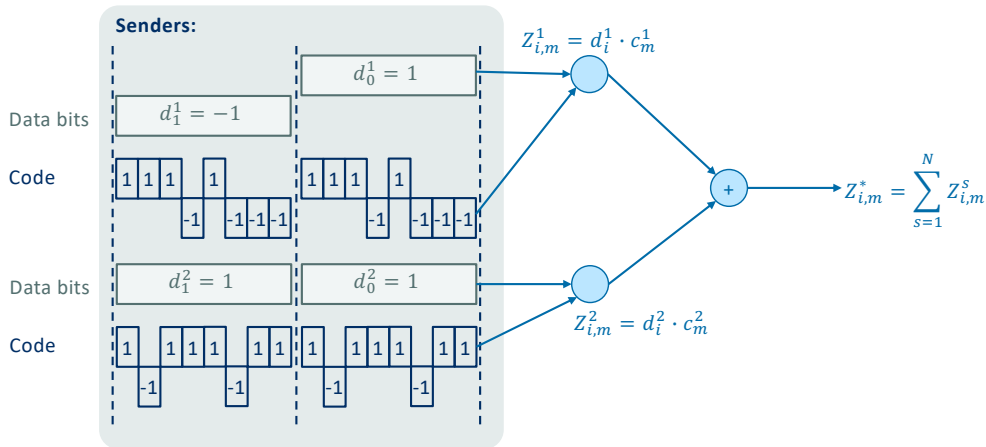
Channel partitioning protocols: CDMA with multiple transmitters



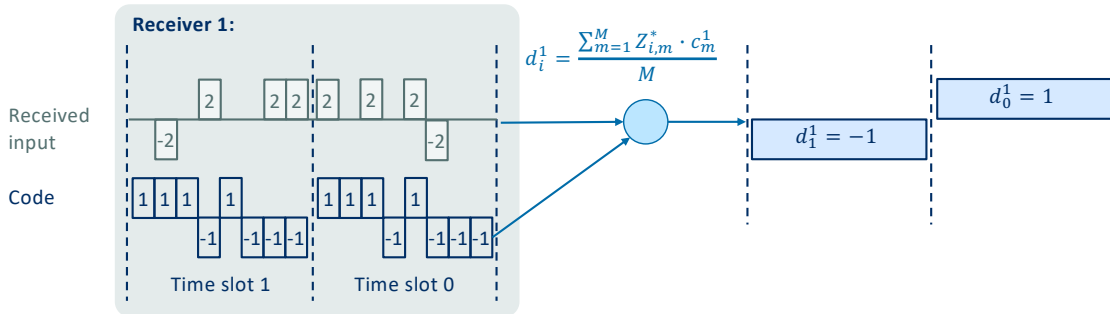
Exercise: CDMA with multiple transmitters



Exercise: Calculate the resulting channel output $Z_{i,m}^*$ for the given example with two transmitters, transmitting 01 and 11 and using codes 11101000 and 10111011 respectively.



Channel partitioning protocols: CDMA receiver with two senders



Assumptions

- CDMA codes must be carefully chosen (i.e., orthogonal codes, such as Walsh-Hadamard, Gold codes, ...)
- Received signal strength from senders is the same (usually not true in practice)
- Transmissions of different users are time-synchronized (not easy in practice)

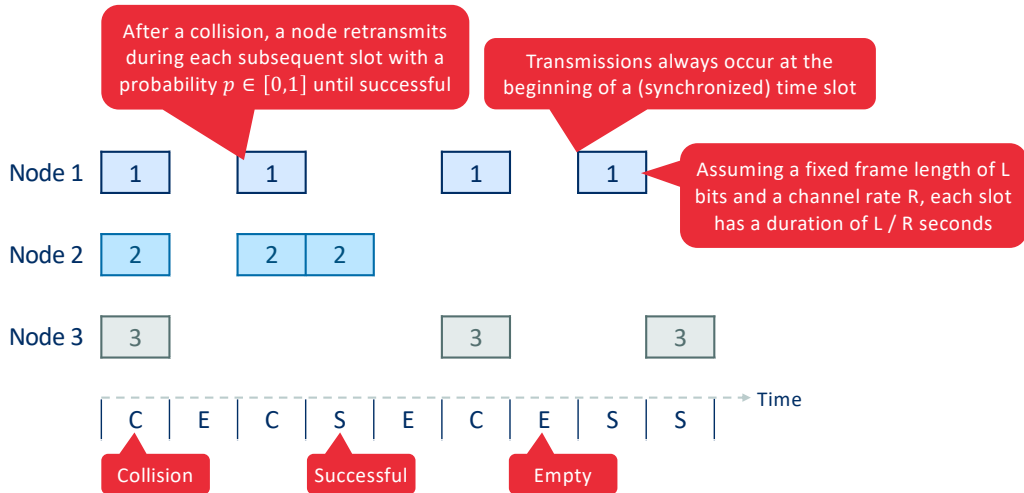
Used in 3G mobile networks

Random access protocols

- Each node transmits at the full rate of the channel
- When there is a collision, nodes repeatedly retransmit their frame
- Each node waits a random delay before retransmitting
- Collisions are avoided if the random delays are different enough

- Well known random access protocols:
 - (slotted) ALOHA
 - Carrier sense multiple access (CSMA), used by Ethernet and Wi-Fi

Random access protocols: Slotted ALOHA



Random access protocols: Slotted ALOHA efficiency

- Assumption: Each node of the set of N nodes transmits during each slot with a probability p
- Probability that a **given node** transmits successfully (i.e., only that node attempts a transmission):

$$p \cdot (1 - p)^{N-1}$$

- Probability that any node transmits successfully:

$$N \cdot p \cdot (1 - p)^{N-1}$$

- The maximum efficiency of slotted ALOHA is achieved by finding p^* , such that:

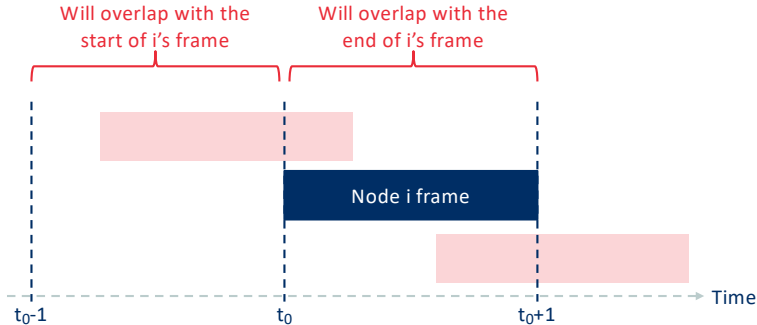
$$p^* = \operatorname{argmax}_p (N \cdot p \cdot (1 - p)^{N-1})$$

i.e., only 37% of the bandwidth is utilized for successful transmissions

- For a large number of senders (i.e., the limit at $N \rightarrow \infty$), the maximum efficiency is $\frac{1}{e} \cong 0.37$

Random access protocols: Pure ALOHA

- Node immediately transmits a frame, instead of waiting for a start of the next time slot
- If a collision occurs, the node immediately retransmits with probability p
- Otherwise, the node waits for a single frame transmission time and retransmits again with probability p



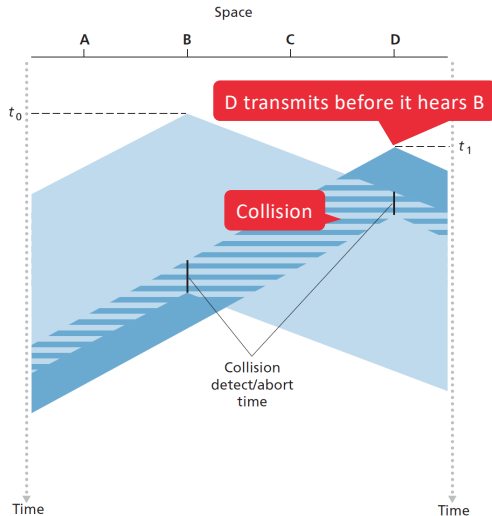
Efficiency of pure ALOHA

- Probability that no other node transmits during $[t_{0-1}, t_0]$:
 $(1 - p)^{N-1}$
- Probability that no other node transmits during $[t_0, t_{0+1}]$:
 $(1 - p)^{N-1}$
- Successful transmission probability:
 $p \cdot (1 - p)^{2(N-1)}$
- Maximum efficiency:
 $\frac{1}{2e} \cong 0.18$

Random access protocols: CSMA/CD

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- Makes use of two important rules to avoid collisions
 - **Carrier sensing:** A node listens to the channel before transmitting, and only transmits if it detects no other ongoing transmissions
 - **Collision detection:** If another transmission is detected while transmitting, stop transmitting and wait a random back-off time before retransmitting
- Collision detection is needed, as even with carrier sensing nodes may still transmit simultaneously



Random access protocols: CSMA/CD random back-off time

- If the back-off time is large and few nodes collide, then the channel will often remain idle
- If the back-off time is small and many nodes collide, then collisions will likely keep occurring
- Optimally, back-off should be long when many nodes collide, and small when few nodes collide
- **Answer: Binary exponential backoff** (used by Ethernet and DOCSIS protocols)
 - After n subsequent collisions, select back-off time of K timeslots uniformly at random from $\{0, 1, 2, \dots, 2^n - 1\}$
 - After a successful transmission, the value of n is set to 0 again
 - Specifically for Ethernet, the actual time equals $K \times 512$ bit times (i.e., K times amount of time needed to transmit 512 bits), with a maximum value for n equal to 10



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Assuming the CSMA/CD protocol in combination with 100 Mbps Ethernet, what is the maximum backoff time assuming the node has failed transmitting a frame 2 times?



0

0 seconds (A)

5.12 μ s (B)

10.24 μ s (C)

15.36 μ s (D)

5.12 ms (E)

10.24 ms (F)

15.36 ms (G)

Random access protocols: CSMA/CD efficiency

- Assumption: A large number of nodes continuously transmitting frames
- Given a maximum propagation delay d_{prop} and a maximum frame transmission time d_{trans} , the efficiency of Ethernet is as follows:

$$\text{Efficiency} = \frac{1}{1 + 5 \cdot d_{prop} / d_{trans}}$$

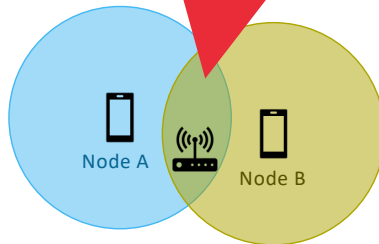
- Examples (for nodes 10 meters away, $d_{prop} = 0.043\mu\text{s}$)
 - 10 Mbps Ethernet ($d_{trans} = 1.21\text{ms}$): Efficiency = 0.9998
 - 100 Mbps Ethernet ($d_{trans} = 121.44\mu\text{s}$) = 0.998
 - 1 Gbps Ethernet ($d_{trans} = 12,14\mu\text{s}$) = 0.98
- Examples (for nodes 2500 meters away, $d_{prop} = 10.75\mu\text{s}$)
 - 10 Mbps Ethernet ($d_{trans} = 1.21\text{ms}$): Efficiency = 0.96
 - 100 Mbps Ethernet ($d_{trans} = 121.44\mu\text{s}$) = 0.69
 - 1 Gbps Ethernet ($d_{trans} = 12,14\mu\text{s}$) = 0.18

Random access protocols: CSMA/CA

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

- Used by Wi-Fi, due to difficulty of performing **collision detection** in wireless networks:
 - In Wi-Fi transmission power is generally much higher than reception power, making it hard (and costly!) to build hardware that can listen while transmitting
 - Due to the hidden terminal problem, not all transmissions can be heard by all nodes in the network
- Wi-Fi uses **collision avoidance** instead
 - If a node has a frame to transmit and the channel is sensed busy, it immediately starts a random backoff counter (using binary exponential backoff)
 - While the channel is busy, the timer is frozen
 - When the counter reaches zero, the frame is transmitted fully and the node waits for an acknowledgment
 - If no acknowledgment is received within a timeout, the maximum backoff is increased and a new random backoff counter is started

Wi-Fi access point can hear A and B, but they cannot hear each other (i.e., they are hidden nodes)

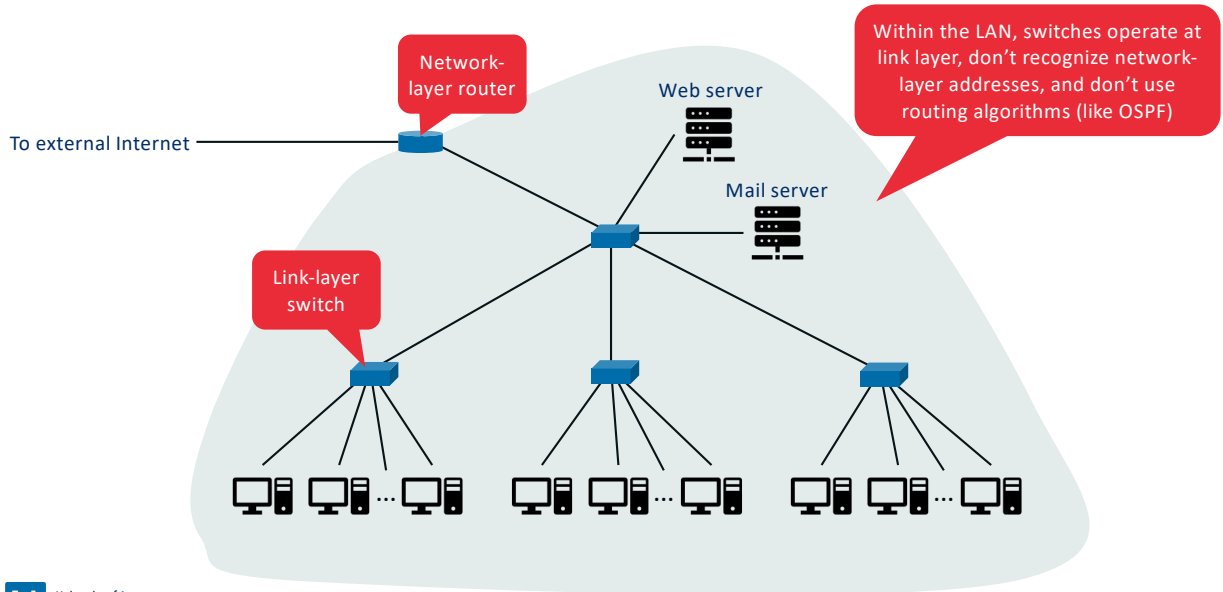


Taking-turns protocols

- Random access protocols (ALOHA, CSMA) do not possess the desirable property that when M nodes are active, their average throughput should be close to R / M bps, which taking-turns protocols do achieve
- **Polling protocol:** A master node polls each node in a round-robin fashion
 - Example: Bluetooth protocol
 - Advantages: Eliminates collisions and empty slots that occur in random access protocols
 - Disadvantages: Polling causes delays, master node results in a single point-of-failure
- **Token-passing protocol:** Only the node holding the token can transmit, and passes the token along after sending a maximum number of frames (or having no more frames left to send)
 - Examples: fibre distributed data interface (FDDI), IEEE 802.5 token ring protocol
 - Advantages: Eliminates collisions and empty slots
 - Disadvantages: Complex failure cases (e.g., a single node failure can crash the channel, recovering from lost tokens, etc.)

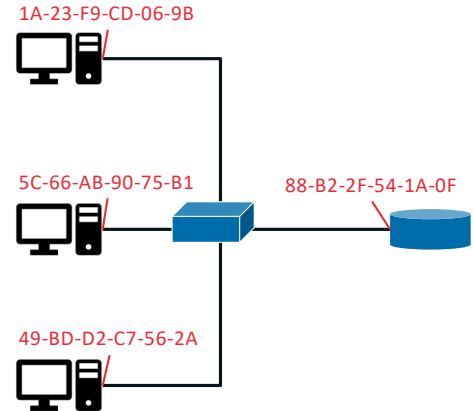
Local Area Networks (LANs)

Switched local area networks (LANs)



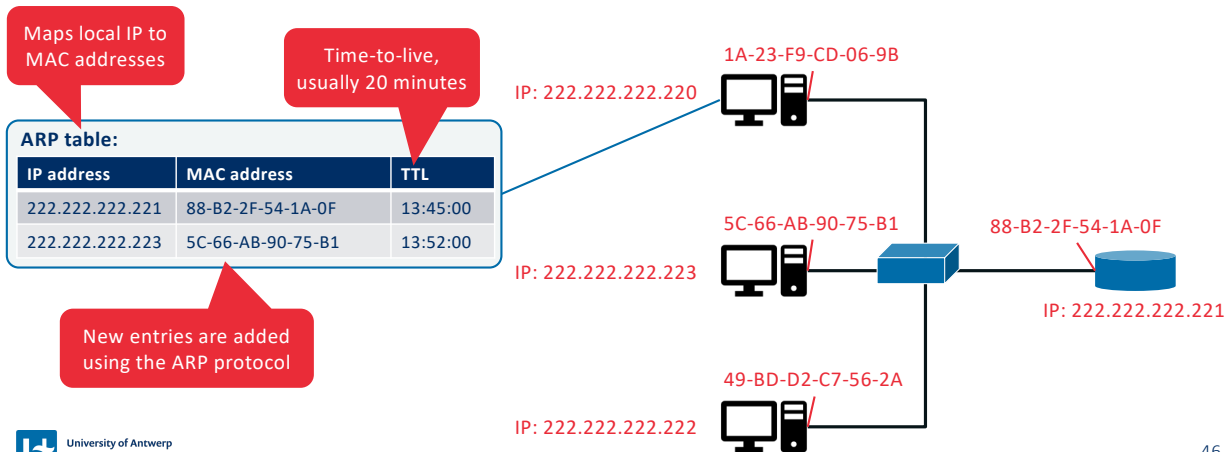
Link-layer addressing

- Addresses at the link layer are called **MAC addresses**
- Each network adapter of a host and router has a MAC address
- Switches do not have MAC addresses (nor IP addresses)!
- Most LAN technologies (e.g., Ethernet, Wi-Fi) use 6-byte (48-bit) MAC address
- Originally MAC addresses were designed to be permanent, but now they can be changed via software
- Each vendor of network adapters is assigned a unique 24-bit MAC address prefix by IEEE, to ensure **all MAC addresses are unique**

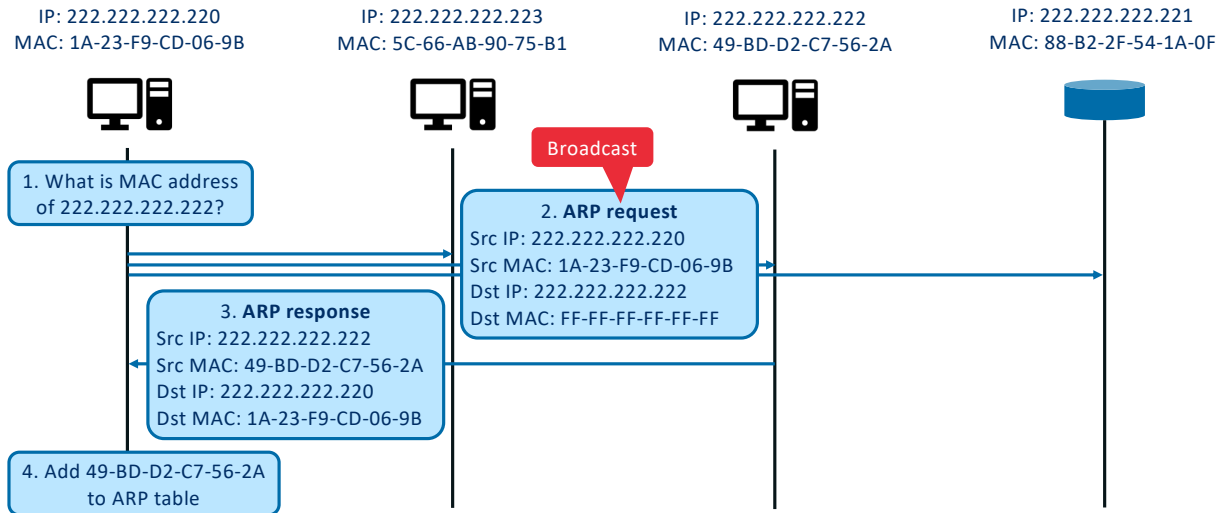


Address Resolution Protocol (ARP) – RFC 826

- Goal: Translate between network-layer address (e.g., IP) and link-layer addresses (i.e., MAC)
- ARP can provide the MAC address associated with an IP address **within the same LAN**

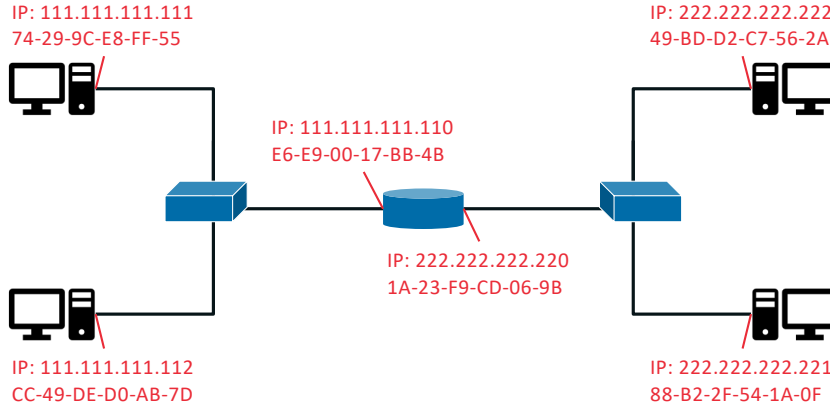


Address resolving using the ARP protocol within a subnet



Sending datagrams across subnets

Src IP: 111.111.111.111
Dst IP: 222.222.222.222
Src MAC: 74-29-9C-E8-FF-55
Dst MAC: ???





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What is the destination MAC address used by the sender in the previous example?



49-BD-D2-C7-56-2A (destination MAC address)

E6-E9-00-17-BB-4B (router MAC address 1)

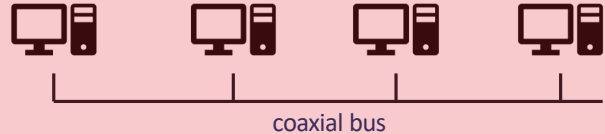
1A-23-F9-CD-06-9B (router MAC address 2)

FF-FF-FF-FF-FF-FF (broadcast address)

The history of Ethernet

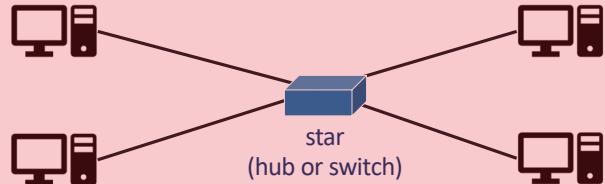
▪ 1970s: Original Ethernet

- Invented by Bob Metcalfe and David Boggs
- Uses coaxial bus to interconnect nodes in broadcast LAN
- CSMA/CD needed to avoid collisions



▪ 1990s: Hub-based star topologies

- Hubs are physical layer devices that forward all incoming bits on all other interfaces
- Also acts as a broadcast LAN (i.e., CSMA/CD required)



▪ 2000s: Switch-based star topologies

- Switches operate at the link layer
- Collision-free store-and-forward packet switching

- Ethernet is **connectionless and unreliable**



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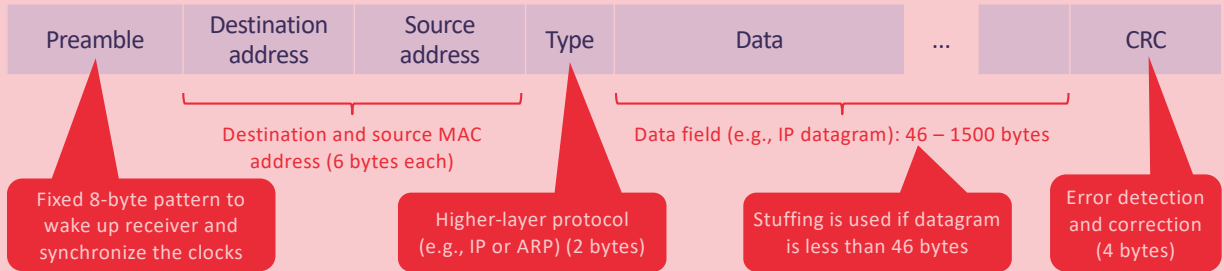
Is CSMA/CD still needed when using a switch-based star topology with Ethernet?

✓✓ 0

Yes

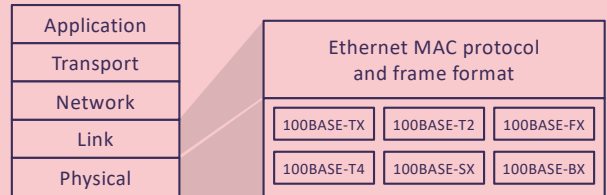
No

Ethernet frame structure



Ethernet technologies

- Ethernet is both a physical and link layer technology
- Many different flavours, identified by acronym consisting of 3 parts (e.g., 100BASE-TX, 10GBASE-SR):
 - Part 1: Data rate (e.g., 10, 100, 1000, 10G, 40G for 10Mbps, 100Mbps, 1Gbps, 10Gbps, 40Gbps data rates)
 - Part 2: BASE, refers to baseband Ethernet (i.e., the physical medium only carries Ethernet traffic)
 - Part 3: Physical media used
 - T for twisted-pair copper
 - F for optical cable
 - S for multi-mode short optical cable
 - X/R for block coding type used
 - ...

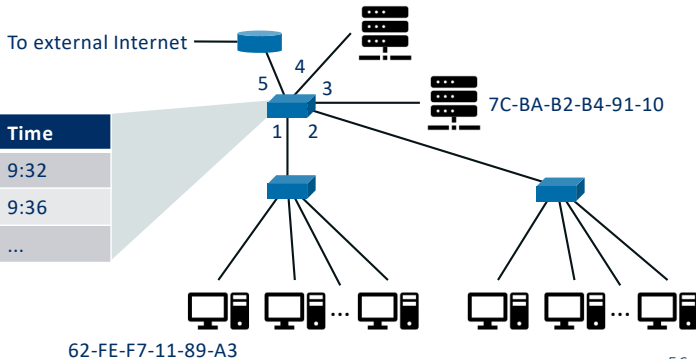


Link-layer switches: Forwarding and filtering

- **Filtering:** Determine whether a frame should be forwarded to some interface or should be dropped
- **Forwarding:** Determine the interfaces to which a frame should be directed (based on switch table)
 - Destination address not found: broadcast frame on all interfaces (except incoming interface)
 - Destination address is linked to incoming interface: filter (discard) frame
 - Destination address is linked to other interface: forward frame to that interface

Switch table:

Address	Interface	Time
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36
...	...	...



Link-layer switches: Self-learning

- Switch tables are built automatically, dynamically and autonomously (i.e., switches are self-learning)
- Steps involved in switch table creation:
 1. The switch table is initially empty
 2. When a frame arrives, the switch stores in its switch table:
 - The MAC address in the frame's source address field
 - The interface from which the frame arrived
 - The current time
 3. If no frames are received from a MAC address within the **aging time**, the address entry is deleted

Switches versus routers

Switches

- Store and forward: Link-layer devices
- Switch table
 - Self-learning based on flooding
 - Uses MAC addresses
- Pros
 - Plug-and-play (no need for manual configuration)
 - High forwarding rate (only process link-layer header)
- Cons
 - Restricted to a spanning-tree topology (to avoid cycles)
 - Switch table grows with number of hosts in subnet
 - Susceptible to broadcast storms (no TTL field)

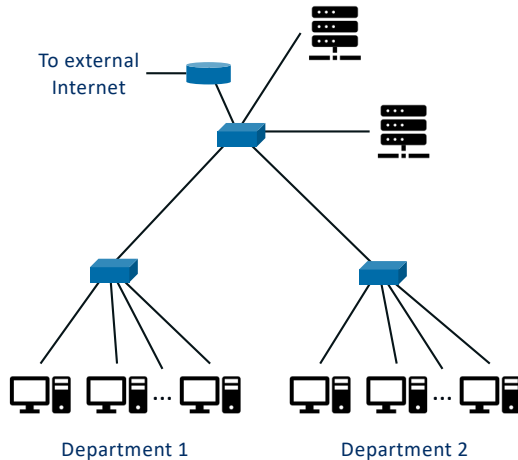
Routers

- Store and forward: Network-layer devices
- Forwarding table:
 - Computed based on routing algorithms
 - Uses IP addresses
- Pros
 - Topology can have redundant paths
 - Firewall protection against layer-2 broadcast storms
- Cons
 - Not plug-and-play (configure IP addresses of connected hosts)
 - Larger processing time per packet (need to process layer-3 fields)

Modern SDN switches (e.g., using OpenFlow) use “match plus action” operations that can make use of both layer-2 frame information and layer-3 datagram information

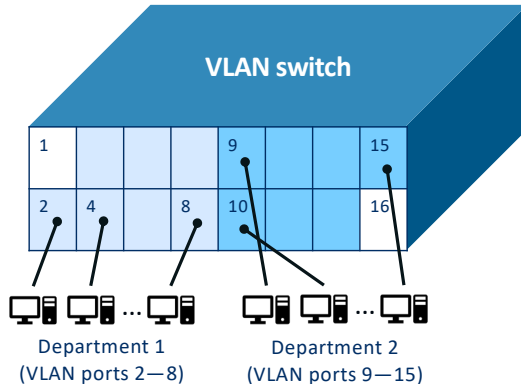
Downsides of switch hierarchies

- LAN topology with hierarchy of switches has several downsides:
 - **Lack of traffic isolation:** Broadcast traffic (ARP, DHCP, frames of unknown destination) traverses entire hierarchy
 - **Inefficient use of switches:** While a large switch could connect all users, it wouldn't provide any traffic isolation
 - **Managing users:** If a users moves between departments, physical cabling needs to be changed
- **Solution:** virtual local area networks (VLANs)!



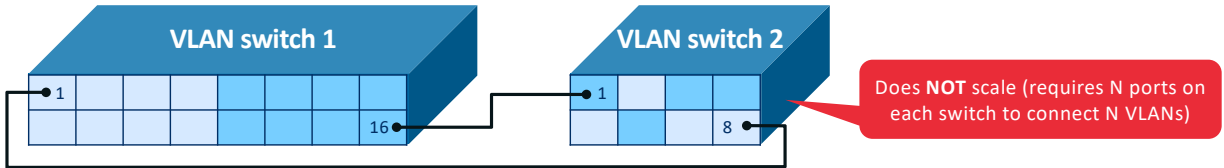
Virtual local area networks (VLANs)

- Multiple *virtual* LANs defined over a single *physical* LAN infrastructure
- In **port-based VLANs**, each switch port (interface) is manually assigned to a VLAN
- Hosts in a VLAN act as if connected to each other via a physical switch
- Each VLAN forms a broadcast domain (i.e., broadcast traffic is not forwarded across VLANs)
- Sending traffic between VLANs can be done by defining a port as belonging to both VLANs and connecting it to a router

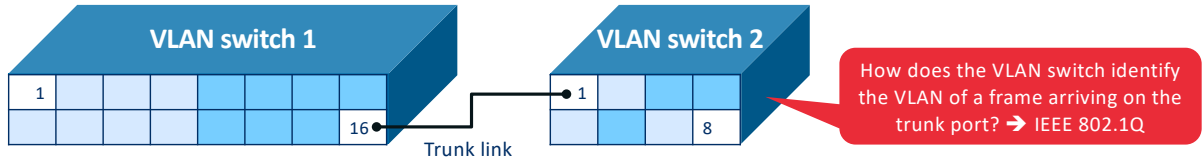


Connecting multiple physical networks in a single VLAN

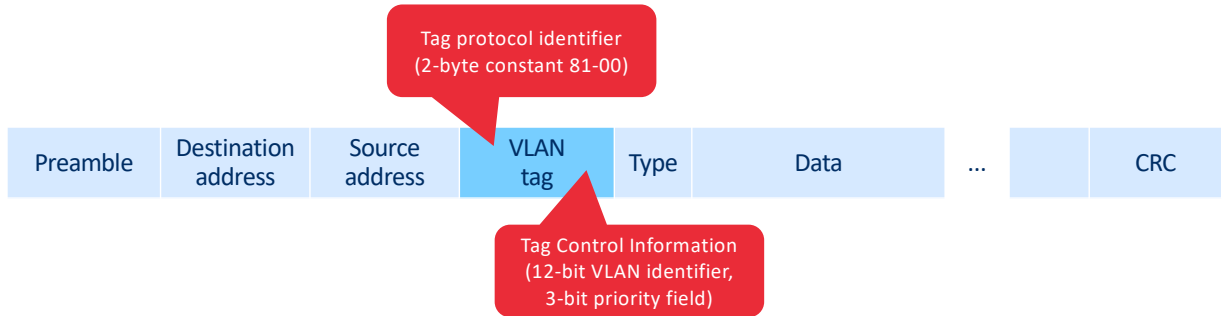
Solution 1: Directly connect port belonging to the same VLAN on both switches with a cable



Solution 2: VLAN trunking (A special port on each switch is configured to interconnect the two VLAN switches)



IEEE 802.1Q: extended Ethernet frame format for VLAN trunking



- Four-byte VLAN tag consists of two-byte tag protocol identifier and two-byte tag control information
- VLAN tag is added to the frame by the sending switch of the VLAN trunk link
- VLAN tag is removed by the receiving switch of the VLAN trunk link

Summary



Summary

- Link-layer is responsible for transferring data across a single link
- **Functionality** can include framing, link access, reliability, and error detection/correction
- Different methods can be used for **error detection**: parity checks, checksums, and cyclic redundancy checks (CRC)
- Multiple access protocols are used to avoid collisions on broadcast links
 - Channel partitioning protocols (FDMA, TDMA, CDMA)
 - Random access protocols (ALOHA, slotted ALOHA, CSMA/CD, CSMA/CA): Used by Ethernet and Wi-Fi
 - Taking-turns protocols (polling, token-passing): Combines advantages of channel partitioning and random access
- Local area networks (LANs) make use of MAC addresses, ARP, and self-learning switches to forward frames
 - VLAN switches can define multiple virtual LANs on a single physical switch for broadcast and traffic isolation
- The contents of this part is covered in the **book** in Sections 6.1, 6.2, 6.3, 6.4, 7.2.1, and 7.3.2

End of Part 5